

PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 4: Data Transformation

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Data transformation

- Why
 - Stabilize variance (e.g., standardization)
 - Meet the assumptions of statistical models (e.g., logit transformation in logistic regression)
 - Make the data more normal distribution-like, improving the validity of association measures like the Pearson's correlation (e.g., Box-Cox transformation)
- Standardization: Rescaling data to have a mean of zero and a standard deviation of one, facilitating comparison across variables of different units.
- Box-Cox (for positive continuous variable only, Box and Cox (1964)):

$$y^{(\lambda)} = \begin{cases} (y^\lambda - 1)/\lambda, & \text{if } y > 0 \text{ \& } \lambda \neq 0, \\ \ln(y), & \text{if } y > 0 \text{ \& } \lambda = 0. \end{cases}$$

- Yeo-Johnson (allowing negative values, Yeo and Johnson (2000)):

$$y^{(\lambda)} = \begin{cases} \{(y+1)^\lambda - 1\}/\lambda, & \text{if } y \geq 0 \text{ \& } \lambda \neq 0, \\ \ln(y+1), & \text{if } y \geq 0 \text{ \& } \lambda = 0, \\ -\{(1-y)^{2-\lambda} - 1\}/(2-\lambda), & \text{if } y < 0 \text{ \& } \lambda \neq 2, \\ -\ln(1-y), & \text{if } y < 0 \text{ \& } \lambda = 2. \end{cases}$$

- Box-Cox with negatives (Hawkins and Weisberg 2017):

$$y^{(\lambda, \gamma)} = \begin{cases} \{(y+\gamma)^\lambda - 1\}/\lambda, & \text{if } y > -\gamma \text{ \& } \lambda \neq 0, \\ \ln(y+\gamma), & \text{if } y > -\gamma \text{ \& } \lambda = 0. \end{cases}$$

- Empirical transformation
 - natural logarithm for economic data
 - logit for percentages

Example: EDA with ISLR2::Wage

Basic understanding of data:

```
?ISLR2::Wage
View(ISLR2::Wage)
colSums(is.na(ISLR2::Wage))
str(ISLR2::Wage)
summary(ISLR2::Wage)
```

Categorizing year and removing region and logwage:

```
library(tidyverse)
wage = ISLR2::Wage |>
  mutate(year = as.factor(year)) |>
  select(-region, -logwage)
summary(wage)
```

Potential Box-Cox transformation on continuous variables:

```
trans_age_wage_1 = car::powerTransform(
  cbind(age, wage) ~ 1, # Box-Cox transformation on age and wage
  family="bcPower", # Box-Cox
  data = wage
)
summary(trans_age_wage_1)
```

```
trans_age_wage_2 = car::powerTransform(
  cbind(age, wage) ~ year + race, # Box-Cox transformation on age and wage adjusting for year and race
  family="bcPower", # Box-Cox
  data = wage
)
summary(trans_age_wage_2)
```

Example: EDA with ISLR::College

Basic understanding of the data:

```
?ISLR2::College
View(ISLR2::College)
str(ISLR2::College)
summary(ISLR2::College)
```

Found unexpected values of Grad.Rate exceeding 100%:

```
library(tidyverse)
ISLR2::College[ISLR2::College$Grad.Rate>100,]
college <- ISLR2::College |>
  mutate(Grad.Rate = ifelse(Grad.Rate > 100, 100, Grad.Rate))
summary(college)
```

Potential Yeo-Johnson transformation on continuous variables:

```
trans = car::powerTransform(
  as.formula(paste0(
    'cbind(',
    paste(names(college)[-1], collapse = ','),
    ')~1'
  )),
  family="yjPower", # Yeo-Johnson
  data = college)
```

)
`summary(trans)`

Bibliography

- Box, G. E. P., and D. R. Cox. 1964. “An Analysis of Transformations.” *Journal of the Royal Statistical Society: Series B (Methodological)* 26: 211–43. <https://doi.org/10.1111/j.2517-6161.1964.tb00553.x>.
- Hawkins, D. M., and S. Weisberg. 2017. “Combining the Box-Cox Power and Generalised Log Transformations to Accommodate Nonpositive Responses in Linear and Mixed-Effects Linear Models.” *South African Statistics Journal* 51: 317–28. <https://doi.org/doi.org/10.37920/sasj.2017.51.2.5>.
- Yeo, In-Kwon, and Richard A. Johnson. 2000. “A New Family of Power Transformations to Improve Normality or Symmetry.” *Biometrika* 87: 954–59. <https://doi.org/10.1093/biomet/87.4.954>.